In [16]:
```
pip install pillow matplotlib
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pillow in c:\programdata\anaconda3\lib\site-packages
(10.4.0)
Requirement already satisfied: matplotlib in c:\programdata\anaconda3\lib\site-packa
ges (3.9.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda3\lib\site
-packages (from matplotlib) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\lib\site-pac
kages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anaconda3\lib\sit
e-packages (from matplotlib) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anaconda3\lib\sit
e-packages (from matplotlib) (1.4.4)
Requirement already satisfied: numpy>=1.23 in c:\users\asus\appdata\roaming\python\p
ython312\site-packages (from matplotlib) (2.4.4)
Requirement already satisfied: packaging>=20.0 in c:\programdata\anaconda3\lib\site-
packages (from matplotlib) (24.1)
Requirement already satisfied: pyparsing>=2.3.1 in c:\programdata\anaconda3\lib\site
-packages (from matplotlib) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\programdata\anaconda3\lib
\site-packages (from matplotlib) (2.9.0.post0)
Collecting numpy>=1.23 (from matplotlib)
  Downloading numpy-1.26.4-cp312-cp312-win_amd64.whl.metadata (61 kB)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-package
s (from python-dateutil>=2.7->matplotlib) (1.16.0)
Downloading numpy-1.26.4-cp312-cp312-win_amd64.whl (15.5 MB)
   ---------------------------------------- 0.0/15.5 MB ? eta -:--:--
   ---- ----------------------------------- 1.8/15.5 MB 11.2 MB/s eta 0:00:02
   ---------- ------------------------------ 4.2/15.5 MB 11.4 MB/s eta 0:00:01
   ---------------- ----------------------- 6.3/15.5 MB 11.0 MB/s eta 0:00:01
   --------------------- ------------------ 8.4/15.5 MB 10.8 MB/s eta 0:00:01
   ---------------------------- ------------ 10.7/15.5 MB 10.6 MB/s eta 0:00:01
   --------------------------------- ------- 12.6/15.5 MB 10.4 MB/s eta 0:00:01
   ------------------------------------- -- 14.7/15.5 MB 10.4 MB/s eta 0:00:01
   ---------------------------------------- 15.5/15.5 MB 9.8 MB/s eta 0:00:00
Installing collected packages: numpy
  Attempting uninstall: numpy
    Found existing installation: numpy 2.4.4
    Uninstalling numpy-2.4.4:
      Successfully uninstalled numpy-2.4.4
Successfully installed numpy-1.26.4
Note: you may need to restart the kernel to use updated packages.
```

```
  WARNING: The script f2py.exe is installed in 'C:\Users\ASUS\AppData\Roaming\Python
\Python312\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning,
use --no-warn-script-location.
ERROR: pip's dependency resolver does not currently take into account all the packag
es that are installed. This behaviour is the source of the following dependency conf
licts.
opencv-python 4.13.0.92 requires numpy>=2; python_version >= "3.9", but you have num
py 1.26.4 which is incompatible.
```

In [19]:
```
from PIL import Image
import numpy as np
import matplotlib.pyplot as plt
```

In [20]:
```
img = Image.open('image.jpg')
print("Image Loaded Successfully")
```

```
Image Loaded Successfully
```

In [21]:
```
img_array = np.array(img)
```

In [22]:
```
print("Shape of image:", img_array.shape)
```

```
Shape of image: (238, 212, 3)
```

In [23]:
```
print("Pixel value at (0,0):", img_array[0,0])
```

```
Pixel value at (0,0): [137 198 253]
```

In [24]:
```
if len(img_array.shape) == 3:
    print("Number of channels:", img_array.shape[2])
else:
    print("Grayscale image (1 channel)")
```

```
Number of channels: 3
```

In [25]:
```
plt.imshow(img)
plt.title("Original Image")
plt.axis('off')
```

Out[25]:
```
(-0.5, 211.5, 237.5, -0.5)
```

Original Image

In [ ]:

```

```